

Performance Testing

Date	04 October 2026
Team ID	POCKETSMART-TEAM
Project Name	PocketSmart AI – Smart Budget & Recommendation Assistant using Gemini Models
Maximum Marks	3 Marks

Testing Scope

PocketSmart AI is a locally run web application. Testing therefore focuses on functional behaviour, validation, authentication, failure handling and local response times. Live Gemini latency depends on Google's service and quota and has not been benchmarked here, so no figure is claimed for it.

Test Areas

Test Area	Scenario	Expected Result	Responsible Member
Registration	New user; duplicate username	Account created; duplicate rejected	Backend Lead
Login	Correct and wrong password	Cookie set on success; wrong password rejected	Backend Lead
Access Control	Protected API and dashboard without sign-in	401 for API; redirect to sign-in for page	Backend Lead
Sign Out	Use token after logout	Token is rejected	Backend Lead
Input Validation	Zero budget; zero guests	Request rejected with a validation error	Testing & Documentation Lead
Planner Generation	Valid home plan request (Gemini mocked)	Result returned and saved to history	AI Integration Lead
History	New user; after a planner call	Empty list; then one entry	Testing & Documentation Lead
AI Failure	Gemini unavailable	Labelled fallback estimate returned, no crash	AI Integration Lead + Testing & Documentation Lead
UI	Desktop and narrow viewport	Forms and results remain readable	Frontend Lead

Automated Test Results

Command: `pytest tests` (from the project folder). Result: 11 passed, 0 failed. Gemini calls are mocked, so the suite runs without an API key.

S.No	Test	Checks
1	<code>test_register_new_user</code>	Successful registration

2	test_register_duplicate_username_rejected	Duplicate username refused
3	test_login_wrong_password_rejected	Wrong password refused
4	test_login_success_sets_cookie	Token and cookie issued
5	test_protected_route_without_token_rejected	401 without token
6	test_dashboard_redirects_without_auth	Redirect to sign-in
7	test_history_empty_for_new_user	Empty history for a new user
8	test_history_populated_after_home_planner_call	Planner call recorded in history
9	test_logout_blacklists_token	Token invalid after sign-out
10	test_home_budget_rejects_non_positive_budget	Budget validation
11	test_party_budget_rejects_zero_guests	Guest-count validation

Measured Local Response Times

Measured with FastAPI TestClient on one development machine (Python 3.12), with the Gemini call forced to fail so the fallback path runs. These figures show application overhead only, not Gemini response time.

Operation	Runs	Median	95th percentile
Register (includes bcrypt hashing)	20	277.8 ms	293.1 ms
Sign in (includes bcrypt check)	10	287.0 ms	288.0 ms
Home planner (fallback)	30	2.1 ms	3.4 ms
Party planner (fallback)	30	2.1 ms	2.7 ms
Jewelry planner (fallback)	30	2.0 ms	2.7 ms
Recommendation history	30	4.4 ms	4.6 ms
Dashboard page	30	1.3 ms	1.5 ms

Registration and sign-in take longer because bcrypt is deliberately slow to protect passwords. For a formal benchmark with Gemini enabled, record request start and end time, AI response time and total response time over repeated requests.

Testing Ownership

Testing & Documentation Lead coordinates test cases and evidence; Backend Lead validates backend behaviour; Frontend Lead checks the UI; AI Integration Lead checks Gemini generation and fallback behaviour.